The Long-Term Effects of Online Channel Adoption in Grocery Retailing

Tanetpong Choungprayoon *   Emelie Fröberg †   Sara Rosengren ‡

Abstract

This study explores the long-term effects of online channel adoption on grocery retailing. Employing the theory of shopping utility maximization, we investigate purchase behaviors of existing customers in categories with different online shopping utility and link them to changes in purchase patterns (shopping frequency and monetary value) and ultimately retailer revenues. Using five-year customer data from a Nordic grocery retailer, we compare (multichannel) customers who adopt an online channel with customers who do not by using propensity score matching and difference-in-differences analysis. The findings show that multichannel customers had their purchasing habits disrupted and this led them to change in the long run; they buy more products in categories with high perceived online ordering convenience and fewer products in categories with high perceived ordering risk. Although they do not change their purchase frequency, the monetary value (per visit) increases, leading to higher total expenditure and thus long-term revenue to the retailer. Our research contributes to the existing literature, which has focused on the short-term effects of online adoption in the grocery context by empirically showing the long-term effects on the category level, customer level, and retailer level, and by providing a systematic bottom-up framework for the retailer to study the effects of customer adoption over time.

Keywords: Multichannel, Grocery retailing; Online adoption; Customer category purchase; Retailer revenue

*Stockholm School of Economics; tanetpong.choungprayoon@hhs.se
†Stockholm School of Economics; emelie.froberg@hhs.se
‡Stockholm School of Economics; sara.rosengren@hhs.se

1 Introduction

Demand for online groceries is increasing and even skyrocketed during the Covid-19 pandemic (Roggeveen and Sethuraman 2020), leading established retailers to invest heavily in multichannel offers where the online channel complements the existing offline channel. A key question for such retailers is how the new online channel will impact the purchase behaviors of existing customers and hence their overall revenue to the retailer, especially in the long term.

Previous studies have shown that channel additions are typically beneficial for retailers, as multichannel customers tend to buy more than do single-channel customers (Ansari, Mela, and Neslin 2008; Kumar and Venkatesan 2005; Venkatesan, Kumar, and Ravishanker 2007). However, this effect holds primarily for hedonic rather than functional product categories (Kushwaha and Shankar 2013), when the new channel helps customers overcome barriers of time and place (Melis et al. 2016) and when it provides different benefits from the established one (Bell, Gallino, and Moreno 2018; Biyalogorsky and Naik 2003; Deleersnyder et al. 2002). What is more, the effect depends on channel experience (Melis et al. 2015; R. J.-H. Wang, Malthouse, and Krishnamurthi 2015), meaning that purchase behaviors might change over time. In fact, in durable and apparel retailing, customers have been shown to revert to their regular purchase patterns after three years (Bilgicer et al. 2015).

However, it is unclear how online channel adoption impacts grocery purchase behaviors in the long term. Grocery differs from other retail sectors due to high-frequency purchases of both hedonic and functional products. In addition, customer habits are typically well-established (Melis et al. 2016; R. J.-H. Wang, Malthouse, and Krishnamurthi 2015) and customers tend to purchase a basket of regularly needed goods in stores that carry basically the same categories (e.g., Vroegrijk, Gijsbrechts, and Campo 2013). Research suggests that purchase behaviors of customers who adopt a new channel (thus becoming multichannel customers) depend on the shopping utility offered by that channel (Chintagunta, Chu, and Cebollada 2012; Vroegrijk, Gijsbrechts, and Campo 2013; Campo and Breugelmans 2015; Campo et al. 2020). Shopping utility comprises both acquisition utility (net benefits offered by the channel in terms of the products offered, e.g., price and assortment) and transaction utility (net benefits offered in terms of how easy it is to acquire the product in a specific channel, e.g., ordering and delivery). Multichannel grocery customers have been found to buy more in product categories with high online ordering and delivery convenience and less in product categories with high perceived purchase risk online, compared to offline shoppers (Melis et al. 2015; Melis et al. 2016; Campo et al. 2020; Campo and Breugelmans 2015). This suggests that transaction utility at the category level is a driver of the changed behaviors. Still, this research has not taken a long-term perspective. Thus, it is not clear to what extent existing customers maintain such new purchasing behaviors (Bilgicer et al. 2015; Avery et al. 2012) or how these behaviors affect overall customer purchase patterns (in terms of frequency of visit and monetary value per visit) and ultimately overall revenue for the retailer. In fact, any short-term increases in purchasing behaviors due to new channel adoption could potentially become new habits in the long run, potentially stabilizing or even attenuating short-term effects.

Employing the theory of shopping utility maximization, this study aims to explore whether the purchase behaviors of existing customers change in the long run after they adopt an online channel. We are particularly interested in the overall effect for the retailer in terms of revenues from existing customers. By investigating purchase behaviors in categories that offer different shopping utility in the online channel and linking those behaviors to customer purchase patterns in terms of (average) purchase frequency and monetary value, we contribute to long-term perspective to the literature. More specifically, our proposed conceptual framework links shopping utility (category effects) to changes in customers' shopping frequency and monetary value (customer effects), and ultimately overall revenue from them (retailer effects).

To empirically study the effects of online channel adoption on existing customers in the long term, this research note utilizes real-world data from a large Nordic grocery retailer, focusing on the online and in-store channels of one of their hypermarkets located in a mid-sized town. This empirical material has three main benefits. First, it includes data on actual purchases, both before and after the introduction of the online channel, thus enabling a quasi-experimental design (where we employ several strategies to overcome the endogeneity issues of self-selection). Second, the dataset follows the same customers for over five years, enabling us to explore the long-term effects. Third, the retailer offered the same assortment, prices, and discounts in both channels, thus allowing us to isolate the effects on the perceived transaction utility. We find that customers who adopted an online channel continue to have a higher total expenditure compared to customers who do not. This suggests that there is a long-term value to multichannel investments for grocery retailers, as the retailer gains more revenue from existing customers who adopt an online channel. Our research contributes to the existing literature by providing a conceptual framework linking the long-term effects on the category level, customer level, and retailer level, and providing initial empirical evidence of these effects.

2 Literature Review

Table 1 offers a summary of the literature on channel addition in retailing. Previous research has investigated the impact of channel addition on different levels, ranging from category to customer to retailer. Of the few studies having investigated the temporal effects, none has focused on the long-term effects (defined as more than two years, Bilgicer et al. 2015) in a grocery retail context.

Several studies have found that customers increase their purchases after adopting a new channel, thus finding beneficial effects for the retailer in terms of revenues from existing customers, an increase in customer shopping frequency and monetary value (Ansari, Mela, and Neslin 2008; Li et al. 2015; R. J.-H. Wang, Malthouse, and Krishnamurthi 2015), share of wallet (Campo et al.

2020; Melis et al. 2016), and profitability (Gensler, Leeflang, and Skiera 2012), In the short run, multichannel customers have higher shopping frequency and monetary value both in the online and offline channels (R. J.-H. Wang, Malthouse, and Krishnamurthi 2015; Campo et al. 2020). They also reallocate their grocery purchase to specific categories (Campo et al. 2020; Campo and Breugelmans 2015). However, Campo and Breugelmans (2015) find that these changes are attenuated by online experience. This means that the shopping utility for multichannel customers may change over time. Still, as previous research has only investigated customer responses in the short term, it is not clear what effects this will have on the overall revenue in the long term. The current study contributes to this literature by focusing on grocery and linking category, customer, and retailer effects in the long term.

Table 1: Overview of Previous Research on the Effects of Channel Addition

Study	**Unit of Analysis**	**Study of Temporal Effects**	**Temporal Term of Effects**	**Context of the Study**	**Dependent Variables**	**Empirical Findings**
Ansari, Mela, and Neslin (2008)	Customer	No	–	Consumer durable and apparel	Shopping frequency, monetary value	Customers migrate to the Internet channel and multichannel users have higher shopping frequency and monetary value
van Nierop et al. (2011)	Customer	No	–	Fashion, sports, living, children's	Shopping frequency, monetary value	Introduction of informational website decreases the customers' shopping frequency and monetary value
Gensler, Leeflang, and Skiera (2012)	Customer, Retailer	No	–	Banking	Revenue, profitability	Adoption of online channel increases revenue and profitability
Avery et al. (2012)	Retailer	Yes	Long-Term	Consumer durable and apparel	Revenue, total customer counts	Introduction of offline store decreases retailers' revenue in the direct channel in the short term but increases their revenue in the long term
Li et al. (2015)	Customer	Yes	Short-Term	Natural health	Shopping frequency, purchasing volume	Late majority and innovators do not purchase more after adopting a new channel; light shoppers do
Bilgicer, Jedidi, Lehmann, and Neslin (2015)	Retailer	Yes	Long-Term	Consumer durable and apparel	Revenues from existing customers	Retailers get more revenues from multichannel customers initially; then the revenues drop as customers return to their regular spending pattern over time
Wang, Malthouse, and Krishnamurthi (2015)	Customer	No	–	Grocery	Shopping frequency, monetary value	Customers increase their shopping frequency and monetary value after adopting a mobile service channel
Melis et al. (2016)	Category, Customer	No	–	Grocery	Share of wallet	Multichannel shoppers have a higher share of wallet at the chain where they started buying online
Campo et al. (2020)	Category, Customer	No	–	Grocery	Share of wallet	Categories that are heavy/bulky and/or can be planned ahead benefit from the online channel. Categories that are sensory and can be planned ahead moderate the effect of price and assortment.
Current study	Category, Customer, Retailer	Yes	Long-term	Grocery	Category purchases, customers' shopping frequency and monetary value, revenues to the retailer	Customers who adopted an online channel had their purchasing habits disrupted and this led them to change in the long term; they buy more product categories with high perceived online ordering convenience and fewer products in categories with high perceived ordering risk. Although existing customers do not change their purchase frequency, the monetary value per visit increases, leading to higher total expenditure and thus long-term revenue to the retailer.

3 Conceptual Framework

In this section, we provide a conceptual framework regarding how customers' online adoption affects their purchase patterns over time and we derive the three research questions that guide our investigation. The framework relies on the theory of shopping utility maximization to explain how customers respond to an online channel. We elaborate on the potential long-term outcomes on existing customers' purchasing habits in specific categories and their revenues to the retailer, which combines customer shopping frequency and monetary value, in the long run (Figure 1).

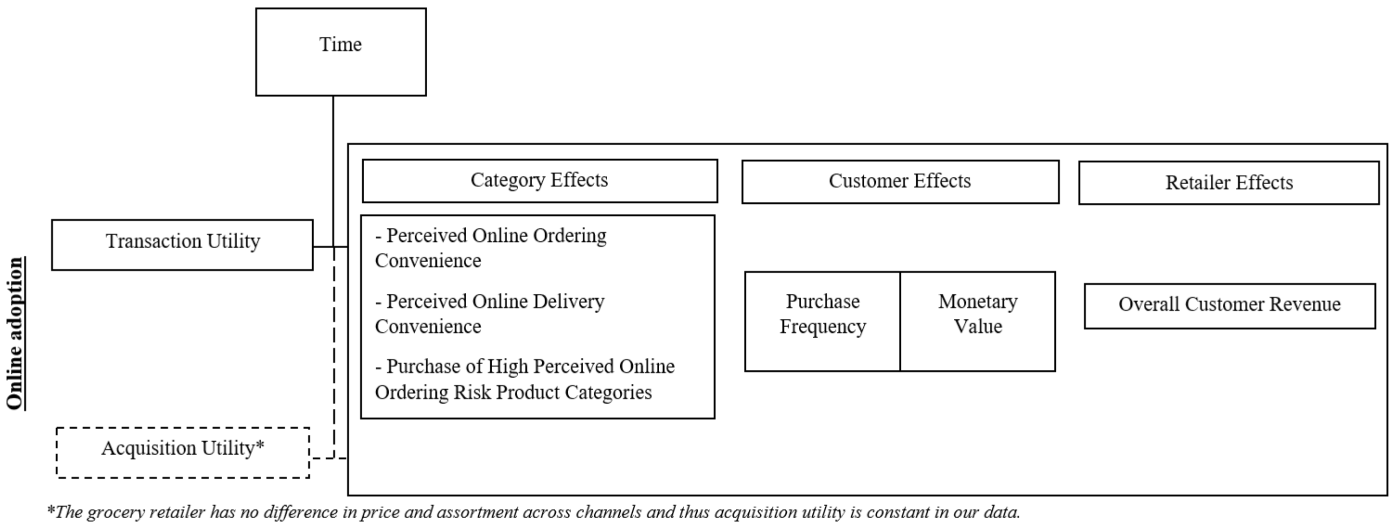

Figure 1: Conceptual Framework

Shopping Utility Maximization

Previous studies on multichannel and multistore purchase behaviors have used shopping utility to explain how customers allocate decisions across those behaviors (Vroegrijk, Gijsbrechts, and Campo 2013; Campo et al. 2020). The theory of shopping utility maximization suggests that when deciding how to purchase groceries across channels, customers can be expected to select a purchase pattern that provides the most overall shopping utility. Purchasing groceries involves the allocation of both money and other scarce recourses such as time and effort. Thus, both monetary and nonmonetary benefits and costs must be considered. Shopping utility can therefore be divided into (a) acquisition utility and (b) transaction utility (Campo et al. 2020), where acquisition utility refers to the net benefits offered by the channel in terms of the product offered (e.g., price, assortment) and transaction utility refers to net benefits offered in terms of how easy it is to acquire the product in a specific channel (e.g., delivery, ordering). Consequently, customers can be expected to decide on the channel in a way that maximizes their total utility, comprising a combination of acquisition and transaction utility.

Shopping utility theory can be used by a retailer to understand what changes will occur for existing customers who adopt the online channel. When the grocery retailer has no difference in price and assortment across channels, acquisition utility is constant and thus the transaction utility for online purchasing will determine the utility of the online channel. Transaction utility varies across different product categories. Previous studies (Campo et al. 2020; Campo and Breugelmans 2015; Melis et al. 2016) show that the online channel enhances transaction utility by offering convenience (time and effort), which in turn leads customers to buy more bulky or heavy product categories (i.e., categories with high perceived ordering convenience) and planned product categories (i.e., categories with high perceived delivery convenience). On the other hand, the online channel also decreases transaction utility in product categories that require physical inspection (i.e., categories with high perceived online purchase risk), leading customers to buy less sensory products online.

Hence, in the short term, we expect to see positive effects on customers' shopping monetary value from high perceived ordering convenience and high delivery convenience product categories and negative effects on existing customers' monetary value from high perceived online purchase risk products (Campo et al. 2020; Campo and Breugelmans 2015; Melis et al. 2016). However, the effects on the shopping frequency and the subsequent revenues to the retailer remain unclear. For example, customers may purchase more high perceived delivery convenience products online while still maintaining their regular purchase of this category offline, and vice versa.

Long-Term Effects on Customer Purchase Behavior and Retailer Revenue

Although the short-term effects of shopping utility maximization have been documented in the literature, the long-term effects on existing customers' shopping behaviors and their total revenue for the retailer are unclear. Online buying experience in grocery shopping can attenuate the category-specific effects caused by online transaction utility (Campo and Breugelmans 2015), meaning that multichannel customers may no longer obtain net (dis)utility from purchasing certain product categories over time. Empirically, Campo and Breugelmans (2015) found that customers purchased a larger share of categories with high perceived online ordering risk and a smaller share of categories with high perceived online delivering convenience as they gained more online buying experience. However, the short-term category-specific effects could also be reinforced over time. This would be due to the net (dis)utility from purchasing specific categories that could (dis)incentivize customers to perform these behaviors repeatedly and turn them into new habits (Shah, Kumar, and Kim 2014).

Theoretically, it is unclear how the attenuating mechanism from aggregate online experience or the reinforcing mechanism from habit formation will impact long-term purchase behaviors. Moreover, the long-term effects for the retailer in terms of revenue will depend on the purchasing patterns developed by the customers in terms of their purchase frequency and the monetary value (per visit) increases. For example, customers who adopted an online channel may maintain their higher level of purchases in categories with high online ordering and high delivery convenience because they enjoy this rewarding behavior, while also increasing their purchases in categories with high online ordering risk due to more experience of obtaining the products over time. As a result, these customers may have higher monetary value but may visit both online and offline channels less, as most of purchases are already made online; hence their total revenues for the retailer remain unchanged. This means that the effects on customers' frequency and monetary value after adopting an online channel over time remain to be explored. Thus, we ask the following research questions about online channel adoption from existing grocery customers:

RQ1: What are the long-term category effects of online channel adoption?

RQ2: What are the long-term customer effects of online channel adoption?

RQ3: What are the long-term retailer effects of online channel adoption?

4 Methodology

In this section, we first describe our data and the empirical setting. Next, we provide initial evidence of the effects of online adoption based on a difference-in-differences (DID) analysis. Accounting for the potential biases from customer self-selection and customer heterogeneity, we employ propensity score matching and estimate DID to investigate the long-term effects on category levels across existing customers, their overall shopping frequency and monetary value, and their revenues to the retailer, respectively.

4.1 Data Description

To evaluate the long-term effects of online adoption, we use real-world data from a Nordic grocery retailer. We use point of sales data from loyalty cards from customers at one specific hypermarket in a mid-sized city. The retailer introduced an online channel in October 2015. At this point, the retailer was the only online grocery offer available in the local market, but in subsequent years several competing options were launched. The hypermarket offers a full range of grocery products both offline and online. Prices and discounts are the same in both channels.

Data come from two different datasets (Oct 2014–Nov 2016 and Aug 2017–July 2019) [1]. We focus on customers that are available in both datasets. As our focus is on channel adoption of existing customers, customers must have made at least 10 purchases before the introduction of the online channel and adoption, and must adopt the online channel within 7 weeks of the introduction to be included in the analysis. In total, we track 578 households (147 of which adopt the online channel) and 116,982 receipts with 1,749,812 products. The retailer charged a picking and delivery fee for the online channel, but this fee is not included in the analyses.

We include a total 358 product categories of which 205 are coded according to perceived online (a) ordering convenience, (b) delivery convenience, and (c) ordering risk.

Table 2 summarizes the measures and definition of variables in this study. All operationalizations are aligned with previous studies. Perceived online ordering convenience, perceived online

[1] We have access to all online/multichannel customers as well as a random sample of 25% of all offline customers.

delivery convenience, and perceived online ordering risk are conceptualized at the category level based on Campo et al. (2021). More specifically, we operationalize this as planning product categories, heavy/bulky product categories, and sensory product categories, respectively.

Table 2: Variables of Interest, Measures, and Definitions

Variables of Interest	**Measures**	**Definition**
Specific Category Purchases		
Perceived online ordering convenience	Planning	Money spent on categories with high perceived online ordering convenience. This includes 205 grocery categories with ratings above 3.5 on a 7-point planned purchase Likert scale (1 = not planned, 7 = usually planned) based on Campo et al. (2021) (local currency).
Perceived online delivery convenience	Heavy/bulky	Money spent on categories with high perceived online delivery convenience. This includes 205 grocery categories with ratings above 3.5 on a 7-point heavy/bulky Likert scale (1 = not heavy/bulky, 7 = very heavy/bulky) based on Campo et al. (2021) (local currency).
Perceived online ordering risk	Sensory	Money spent on categories with high perceived online ordering risk. This includes 205 grocery categories with ratings above 3.5 on a 7-point sensory Likert scale (1 = not sensory, 7 = very sensory) based on Campo et al. (2021) (local currency).
Overall Shopping Behaviors		
Frequency	Frequency	Average number of purchase incidences per week (times).
Monetary value	Monetary	Average amount of money spent per purchase incidence (local currency).
Total Customer Revenues		
Total expenditures	Expenditure	Total amount of money spent by customers over a given period (local currency).
Period of Time		
Before adopting an online channel	Pre	52 weeks prior to the first online purchase of the treated household.
After adopting an online channel (short term)	Short	0–52 weeks following the first online purchase.
Over 2 years after adopting an online channel	Long	146–198 weeks following online introduction.

4.2 Model-Free Evidence

To provide initial evidence of the effects of online adoption, we first implemented a simple DID analysis comparing customers who adopted an online channel (Multichannel) and those who did not (Offline). In Table 3, the DID columns represent the isolated difference compared to the average pre-period and online adoption effects (see Figure A1 in Appendix for parallel trend assumption). We assessed the first difference in category purchases, customers' shopping frequency and monetary value, and their total revenues to the retailer for the year prior to their online channel adoption (*Pre*), and relative differences in performance up to four years after the adoption (short-term (*Short*), and long-term (*Long*), respectively). This model-free evidence shows that the existing customers who adopt the online channel significantly change their category purchase patterns as well as their overall frequency and monetary value in the long run, leading to higher revenue for the retailer. However, the model-free results do not account for the self-selection bias of those existing customers who decide to adopt an online channel (i.e., multichannel shoppers are likely to shop significantly more than offline shoppers, which shows that customers with a higher shopping frequency and monetary value are likely to adopt an online channel (Boehm 2008; Konuş, Verhoef, and Neslin 2008; Kumar and Venkatesan 2005). Hence, it is difficult to conclude whether this long-term change is due to the online adoption or to self-selection bias.

Table 3: Model-Free Evidence of the Effects of Online Channel Adoption

		Pre	**Short**	**DID**	**Long**	**DID**
Planning	Multichannel	1627	1480	-366	1441	381
	Offline	1242	1462			675
Heavy/Bulky	Multichannel	29	53	0	48	23
	Offline	29	53			26
Sensory	Multichannel	388	445	25	113	-136
	Offline	201	233			62
Frequency	Multichannel	1.19	1.23	0.07	1.16	0.09
	Offline	0.93	0.90			0.80
Monetary Value	Multichannel	622	661	49	706	83
	Offline	339	329			340
Expenditure	Multichannel	32440	36526	4873	35263	4193
	Offline	15935	15147			14565

4.3 Propensity Score Matching

Since customers self-select to the online channel, it may be that customers who adopt the online channel are genuinely different from those who do not, regardless of online channel adoption. To deal with this issue, we employ propensity score matching, allowing us to match multichannel customers with offline customers who are similar to them prior to channel adoption. We argue that any differences between customers adopting online channel and those who do not should be reflected in pre-introduction purchasing behaviors, captured by frequency and monetary value, but also include the distance to the offline store in our matching. Modeling the ith customer's online adoption in period t as the dependent variable, we estimate propensity scores using the following regression model:

$$\text{Treat}_{i,post} = \alpha_0 + \alpha_1 \text{Frequency}_{i,pre} + \alpha_2 \text{MonetaryValue}_{i,pre} + \alpha_3 \text{Distant}_{i,pre} + \varepsilon_i \tag{1}$$

where post is a period of seven weeks after the introduction of the online channel, pre is one year prior to the introduction of the online channel, and Distant is a dummy variable that takes the value 1 if the customer's registered postal code is more than three kilometers from the physical store.

Upon matching, our samples are more balanced (see Appendix A1 for summary statistics before and after matching). We further explore additional characteristic that could be influenced by channel adoption, namely the number of units purchased (*Volume*), deal proneness represented by the ratio of products purchased on deals to total purchase volume (*Deal*), and the family situation reflected in total number of purchased diapers (*Family*) (Kumar and Venkatesan 2005; Venkatesan, Kumar, and Ravishanker 2007; Chu, Chintagunta, and Cebollada 2008; Chu et al. 2010; Kushwaha and Shankar 2013). As there is still some variation in terms of *Volume*, *Deal*, and *Family* between matched pair i in the matching period, we include these variables as controls in our DID Analysis.

4.4 DID Analysis

We calculate a difference in all our dependent variables between the matched pairs. We then test the following regression equation,

$$Y_{i,t} = \beta_1 \text{Short}_{i,t} + \beta_2 \text{Long}_{i,t} + \beta_3 \text{Volume}_{i,t} + \beta_4 \text{Deal}_{i,t} + \beta_5 \text{Family}_{i,t} + \xi_i + \varepsilon_i \tag{2}$$

where $Y_{i,t}$ is the difference between the *Multichannel* and *Offline* matched pair i in period t of the customers' purchase behaviors (*Planning*, *Heavy/Bulky*, *Sensory*, *Frequency*, *Monetary Value*, and *Expenditures*). Consistent with the model-free evidence, purchase behaviors in the year prior to online channel adoption, and relative differences in category purchases, customers' shopping frequency and monetary value, and their total revenues to the retailer up to four years after the adoption are investigated. Period t includes the period before the first purchase (*Pre*), the short-term (*Short*), and the long-term (*Long*). *Short* and *Long* are dummy variables.

As we are interested in long-term effects, we focus on the β_2 coefficient (pre is a baseline situation when β_1 and β_2 are equal to zero). We use a fixed effect model to account for unobserved heterogeneity. Thus, our model tests the pairwise difference between customers who adopted the online channel and customers who did not, and the difference between the period prior to adoption and the long-term period following adoption. Given that we take a full year into account, we do not control for seasonal effects.

5 Results

Table 4 shows the results of our DID analysis of pair-wise matches between *Multichannel* and *Offline* customers. In the Appendix (Figure A2), we illustrate that the parallel trend assumption holds.

At the category level, the results indicate that the short-term effects from changes in transaction utility found in previous research are both attenuated by online buying experience and reinforced by habit formation in the long run. First, there is no difference between the purchases of products with high perceived online ordering convenience (*Planning*) in the long-term period and the period prior to adoption, between the multichannel and the offline customers, 95% CI [-681, 694.1]. Second, purchases of products with high perceived online delivery convenience (*Heavy/Bulky*) are higher in the long-term period, 95% CI [104.4, 572.19]. This category purchases are higher both online and offline for the multichannel customers, indicating that they enjoy purchasing not only from the online channel that provides a direct increase in shopping utility, but also from the offline channel. Surprisingly, the purchases of products with high perceived online ordering risk (*Sensory*) remain lower in the long-term period, 95% CI [-788.84, 188.09]. This suggests a reinforcing effect in which multichannel customers become used to purchasing less in this category over time. Answering RQ1, we thus find that, in the long term, existing customers who adopted the online channel purchase more in product categories with high perceived online delivery convenience, but less in product categories with high ordering risk.

At the customer level, the results show that multichannel customers make fewer frequent purchases (*Frequency*), 95% CI [-0.22, 0], but in the long term spend a higher monetary value per purchase occasion (*Monetary Value*), 95% CI [54.71, 152.41], compared to before adoption. Although this reflects a reversion in customers' shopping patterns in terms of frequency (i.e., they visit the same number of times regardless of the channel), they buy more, especially from high perceived online delivery convenience categories. Answering RQ2, this suggests a stable purchase pattern in terms of frequency, but a higher monetary value for existing customers in the long term.

At the retailer level, the results suggest that multichannel customers have higher total expenditures (*Expenditure*) in the long-term period compared to the period prior to adoption, 95% CI [-356.05, 1488.53]. Answering RQ3, this implies a higher total revenue from existing customers adopting the online channel for the grocery retailer in the long term. This increase in revenues is due to the long-term increase in overall monetary value of multichannel customers after adopting an online channel, driven by categories with high perceived online ordering convenience.

Table 4: Fixed Effect Panel Regression Results of the DID Analysis of Pair-Wise Matches between Multichannel and Offline Customers

	Category Effects			**Customer Effects**		**Retailer Effects**
	Planning	Heavy/Bulky	Sensory	Frequency	Monetary	Expenditure
Panel A: Multichannel Customer Purchases (Total)						
Short	1114.53 (337.15)	247.33 (114.69)	570.62 (239.52)	-0.08 (0.05)	45.63 (23.96)	643.20 (452.25)
Long	6.55 (343.77)	338.29 (116.95)	-300.38 (244.23)	-0.11 (0.06)	103.56 (24.43)	566.24 (461.15)
R-squared	0.88	0.73	0.69	0.41	0.23	0.93
Panel B: Multichannel Customer Purchases (Online Channel)						
Short	1156.47 (359.30)	624.51 (122.11)	-71.11 (231.30)	-0.33 (0.07)	431.73 (57.71)	703.20 (438.89)
Long	695.39 (347.54)	287.13 (118.11)	245.72 (223.73)	-0.35 (0.06)	307.47 (55.82)	-1842.49 (424.53)
R-squared	0.93	0.80	0.88	0.69	0.23	0.97
Panel C: Multichannel Customer Purchases (Offline Channel)						
Short	-197.23 (312.31)	244.98 (112.28)	-138.20 (225.56)	0.03 (0.05)	-24.38 (23.85)	363.39 (474.01)
Long	-650.46 (309.30)	129.50 (111.20)	-309.66 (223.39)	-0.02 (0.05)	25.61 (23.62)	119.43 (469.43)
R-squared	0.87	0.66	0.69	0.45	0.15	0.92

Notes: Standard errors in parentheses; includes covariates specified in Eq. 2.

6 Discussion

This investigation sets out to provide retailers with a better understanding of how online channel adoption impacts the purchase behaviors of existing customers and hence their overall revenue, especially in the long term. Our results show that customers who adopt an online channel had their purchasing habits disrupted, but also that their habits changed in the long term. We find that existing customers who adopt an online channel continue to purchase more categories with high perceived delivery convenience while avoiding categories with high perceived ordering risk over time. However, they do not appear to purchase more products of high ordering convenience—that is, their purchases do not become more planned—in the long term. Moreover, existing customers who adopt an online channel purchase more (per visit), while their frequency of visits remains the same. Thus, the purchase habits are stable in terms of frequency and the category effects mainly influence the monetary value. This is in line with the insignificant effect on ordering convenience in which customers do not make more planned purchases in the long term. Overall, these changes lead to higher total expenditure and thus to more revenues for the retailer.

Our findings contribute to previous research by showing the persistent, long-term effects on various aspects of customer purchasing habits. However, it should be noted that our empirical results hint an attenuating effect of online adoption over time—the magnitudes of the effects are smaller in the long term compared to the short term. In the short term, the results are consistent with the existing literature, in which an online channel increases transaction utility for online purchases, inducing customers to buy more products with high online ordering convenience and high delivery convenience from an online channel (Campo et al. 2020; Campo and Breugelmans 2015) and leading customers to spend more (higher monetary value) and buy more often (frequency; Wang, Malthouse, and Krishnamurthi 2015). However, contrary to these studies, we also found that customers purchase more product categories with high online ordering risk, although this positive effect decreases over time. One possibility is that the customers who adopt the online channel early are less risk-averse overall. To explore this, we re-ran the analysis on customers that adopted the online channel three years after the channel was introduced and found that these adopters indeed purchased fewer products with high perceived risk upon adopting the online channel (see more in Appendix Table A2). These results suggest that retailers must be careful when interpreting short-term purchase behaviors of early adopters, as they are less likely to indicate what other customer groups will do, especially in the long term.

The proposed framework links shopping utility (category level), to changes in purchase patterns in terms of shopping frequency and monetary value (customer level), and ultimately overall revenue (retailer level). This framework can be used by retailers to better understand the bottom-up effect of offering a new channel starting with utility at the category level. By using this framework when analyzing point of sales data from loyalty cards, retailers can follow changes in customer behavior at all three levels across each channel as well as in total. For our grocery retailer, the transaction utility offered by the new online channel had a positive impact on the purchase behaviors of existing customers and hence their overall long-term revenue to the retailer. For other retailers, the relationship could differ. This is especially true for retailers that use different pricing and assortments across channels. For such retailers, our framework can be extended to also study acquisition utility.

Overall, our results present initial evidence of the benefits to retailers of offering several channels to existing customers in grocery retail, although they remain limited by the specific strategies of the retailer in our study. Future research is needed to investigate the generalizability of our findings. Such research should also explore how acquisition utility impacts the purchase behaviors of existing customers in the long term. Similarly, the framework could be used to provide more in-depth insights into channel adoption in other retail categories, which to date has mainly focused on customer-level effects (Avery et al. 2012; Bilgicer, Jedidi, Lehmann, and Neslin 2015). It is our hope that this study will help spur additional research on this managerially relevant topic.

References

Ansari, Asim, Carl F. Mela, and Scott A. Neslin (2008). "Customer Channel Migration." *Journal of Marketing Research*, 45 (1), 60–75.

Avery, Jill, Thomas J. Steenburgh, John Deighton, and Mary Caravella (2012). "Adding Bricks to Clicks: Predicting the Patterns of Cross-Channel Elasticities over Time." *Journal of Marketing*, 76 (3), 96–111.

Bell, David R., Santiago Gallino, and Antonio Moreno (2018). "Offline Showrooms in Omnichannel Retail: Demand and Operational Benefits." *Management Science*, 64 (4), 1629–1651.

Bilgicer, Tolga, Kamel Jedidi, Donald R. Lehmann, and Scott A. Neslin (2015). "The Long-Term Effect of Multichannel Usage on Sales." *Customer Needs and Solutions*, 2 (1), 41–56.

Biyalogorsky, Eyal and Prasad Naik (2003). "Clicks and Mortar: The Effect of On-Line Activities on Off-Line Sales." *Marketing Letters*, 14 (1), 21–32.

Boehm, Martin (2008). "Determining the Impact of Internet Channel Use on a Customer's Lifetime." *Journal of Interactive Marketing*, 22 (3), 2–22.

Campo, Katia and Els Breugelmans (2015). "Buying Groceries in Brick and Click Stores: Category Allocation Decisions and the Moderating Effect of Online Buying Experience." *Journal of Interactive Marketing*, 31, 63–78.

Campo, Katia, Lien Lamey, Els Breugelmans, and Kristina Melis (2020). "Going Online for Groceries: Drivers of Category-Level Share of Wallet Expansion." *Journal of Retailing*.

Chintagunta, Pradeep K., Junhong Chu, and Javier Cebollada (2012). "Quantifying Transaction Costs in Online/Offline Grocery Channel Choice." *Marketing Science*, 31 (1), 96–114.

Chu, Junhong, Marta Arce-Urriza, Jose-Javier Cebollada-Calvo, and Pradeep K. Chintagunta (2010). "An Empirical Analysis of Shopping Behavior Across Online and Offline Channels for Grocery Products." *Journal of Interactive Marketing*, 24 (4), 251–268.

Chu, Junhong, Pradeep Chintagunta, and Javier Cebollada (2008). "A Comparison of Within-Household Price Sensitivity Across Online and Offline Channels." *Marketing Science*, 27 (2), 283–299.

Deleersnyder, Barbara, Inge Geyskens, Katrijn Gielens, and Marnik G. Dekimpe (2002). "How Cannibalistic Is the Internet Channel?" *International Journal of Research in Marketing*.

Gensler, Sonja, Peter Leeflang, and Bernd Skiera (2012). "Impact of Online Channel Use on Customer Revenues and Costs to Serve." *International Journal of Research in Marketing*, 29 (2), 192–201.

Konus, Umut, Peter C. Verhoef, and Scott A. Neslin (2008). "Multichannel Shopper Segments and Their Covariates." *Journal of Retailing*, 84 (4), 398–413.

Kumar, V. and Rajkumar Venkatesan (2005). "Who Are the Multichannel Shoppers and How Do They Perform?" *Journal of Interactive Marketing*, 19 (2), 44–62.

Kushwaha, Tarun and Venkatesh Shankar (2013). "Are Multichannel Customers Really More Valuable?" *Journal of Marketing*, 77 (4), 67–85.

Li, Jing, Umut Konus, Koen Pauwels, and Fred Langerak (2015). "The Hare and the Tortoise: Do

Earlier Adopters of Online Channels Purchase More?" *Journal of Retailing*, 91 (2), 289–308.

Melis, Kristina, Katia Campo, Els Breugelmans, and Lien Lamey (2015). "The Impact of the Multi-Channel Retail Mix on Online Store Choice." *Journal of Retailing*, 91 (2), 272–288.

Melis, Kristina, Katia Campo, Els Breugelmans, and Lien Lamey (2016). "A Bigger Slice of the Multichannel Grocery Pie." *Journal of Retailing*, 92 (3), 268–286.

Roggeveen, Anne L. and Raj Sethuraman (2020). "How the COVID-19 Pandemic May Change the World of Retailing." *Journal of Retailing*, 96 (2), 169.

Shah, Denish, V. Kumar, and Kihyun Hannah Kim (2014). "Managing Customer Profits: The Power of Habits." *Journal of Marketing Research*, 51 (6), 726–741.

Venkatesan, Rajkumar, V. Kumar, and Nalini Ravishanker (2007). "Multichannel Shopping: Causes and Consequences." *Journal of Marketing*, 71 (2), 114–132.

Vroegrijk, Mark, Els Gijsbrechts, and Katia Campo (2013). "Close Encounter with the Hard Discounter." *Journal of Marketing Research*, 50 (5), 606–626.

Wang, Kitty and Avi Goldfarb (2017). "Can Offline Stores Drive Online Sales?" *Journal of Marketing Research*, 54 (5), 706–719.

Wang, Rebecca Jen-Hui, Edward C. Malthouse, and Lakshman Krishnamurthi (2015). "On the Go: How Mobile Shopping Affects Customer Purchase Behavior." *Journal of Retailing*, 91 (2), 217–234.

Appendix

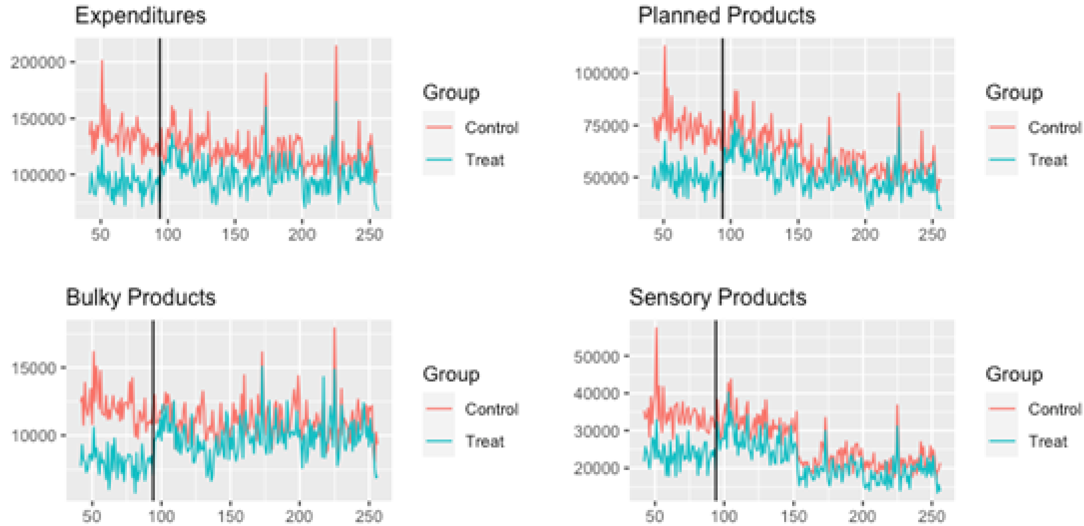

Figure A1: Purchasing behaviors over time of the treatment group, which represents customers who adopted an online channel (Multichannel), and control group, which represents customers who never adopted an online channel (Offline) without adjusting for self-selection bias

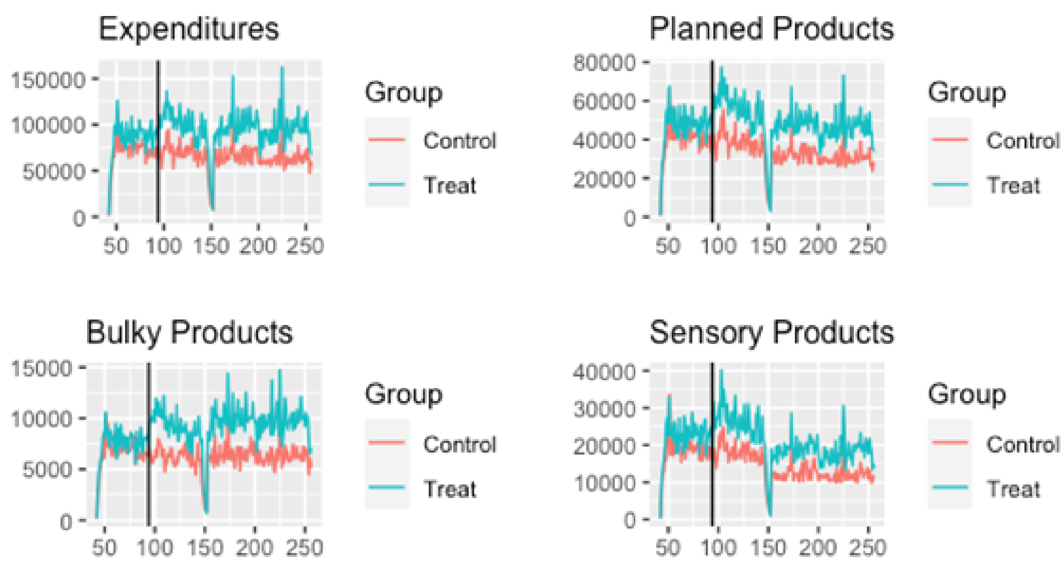

Figure A2: Purchasing behaviors over time of the treatment group, which represents customers who adopted an online channel (Multichannel), and control group, which represents customers who never adopted an online channel (Offline) after adjusting for self-selection bias

Table A1: Summary Statistics for All Samples and Comparison Before and After Matching

	Multichannel	**Offline (Matched)**	**Offline (Unmatched)**	**All Customers**
Frequency	1.20 (1.12)	1.18 (0.89)	0.93 (0.75)	0.69 (0.77)
Monetary	609 (278)	520.91 (254)	339.64 (218)	369.72 (268)
Distant	0.23 (0.42)	0.23 (0.42)	0.45 (0.50)	0.40 (0.49)
Volume	1494 (1013)	1301.47 (749)	799.68 (688)	661.13 (768)
Deals (% of volume)	0.37 (0.39)	0.43 (0.45)	0.56 (0.60)	1.61 (5.46)
Family	8.93 (13.90)	2.32 (8.02)	1.26 (5.61)	1.99 (6.83)
N	146	146	431	4674

Notes: M (SD)

Table A2: Fixed Effect Analysis of Late Adopters

	Planning	Bulky	Sensory	Frequency	Monetary	Total Expenditures
Panel A: Multichannel Customer Purchases (Total)						
Short	361.23 (388.03)	316.18 (142.49)	-275.10 (211.01)	-0.02 (0.07)	96.31 (21.58)	689.59 (532.89)
R-squared	0.95	0.78	0.93	0.76	0.19	0.97
Panel B: Multichannel Customer Purchases (Online Channel)						
Short	782.34 (547.58)	169.56 (179.46)	533.21 (301.25)	0.00 (0.10)	383.60 (58.70)	-972.98 (719.42)
R-squared	0.96	0.86	0.94	0.84	0.39	0.98
Panel C: Multichannel Customer Purchases (Offline Channel)						
Short	141.40 (371.49)	195.05 (131.72)	-78.98 (218.35)	0.09 (0.07)	20.12 (21.24)	412.76 (511.04)
R-squared	0.95	0.81	0.93	0.77	0.03	0.97

Notes: In this analysis, August 1, 2018 is used as the late treatment date. Propensity score matching is performed on the 52 weeks prior to that date. Customers are allowed to adopt the online channel within ±4 weeks of the late treatment date, resulting in 95 matched pairs of adopters and control customers. The analysis compares 48 weeks before (pre) and 48 weeks after (short). The model specification follows the main analysis, but long-term effects are not available.